Take-Home Project
Census Income Classification and Customer Segmentation Report

Executive Summary

This project addressed two related marketing questions using the provided census dataset. The first objective was to predict whether an individual belongs to the income group earning more than $50,000 or at or below $50,000. The second objective was to build a segmentation model that groups people into meaningful customer profiles for marketing use.

For the classification task, I compared Logistic Regression and Random Forest after applying a consistent preprocessing pipeline for mixed numeric and categorical features. Both models performed well on a held-out test set, with overall accuracy near 95% and ROC-AUC above 0.94. The models were very close in performance, so I recommend Logistic Regression as the primary model because it offers nearly the same predictive quality while being easier to explain and maintain.

For the segmentation task, I used K-Means clustering after preprocessing the data and reducing dimensionality with Principal Component Analysis (PCA). I compared solutions from 2 to 8 clusters using silhouette score. The best score among the tested solutions was achieved at 7 clusters. That solution produced several clearly distinct groups, including children or dependents, older non-working adults, and a small affluent segment with very high investment income. At the same time, some of the seven clusters were fairly similar in practical business terms. My recommendation is to retain the 7-cluster solution as the analytical model, while considering a smaller set of operational marketing personas if the client wants a simpler campaign structure.

Taken together, the two models provide complementary value. The classifier helps prioritize likely higher income individuals, while the segmentation model helps tailor offers and messaging based on the broader profile of each group.

1 Business Objective

The client's first goal was to distinguish between two groups of people for marketing purposes: those earning more than $50,000 and those earning $50,000 or less. This is a supervised learning problem because the dataset includes a known income label for each record.

The second goal was to create a segmentation model of the same population for marketing use. This is an unsupervised learning problem. Instead of predicting a known label, the aim is to discover natural groups in the data and then describe how those groups differ in a way that can support campaign planning.

These two goals are related but serve different business purposes. The classifier supports targeting efficiency, while the segmentation model supports customer understanding and message design.

2 Dataset Overview

The dataset contains weighted census records extracted from the 1994 and 1995 Current Population Surveys. Each record includes 40 demographic and employment-related variables, plus a survey weight and an income label.

After loading the provided files, the working dataset contained 199,523 rows and 42 columns. The features included a mix of numeric variables such as age, wage per hour, capital gains, capital losses, dividends from stocks, and weeks worked in year, along with categorical variables such as class of worker, education, marital status, race, sex, and occupation-related fields.

The target variable was highly imbalanced. Most records belonged to the lower-income group, while the greater-than-$50,000 group represented a much smaller share of the dataset. This mattered when evaluating model quality because accuracy alone would have been misleading.

The dataset also includes a weight column. Based on the project description, this weight represents how much of the general population each row stands for due to stratified sampling. I therefore treated it as meaningful survey information rather than just another input field.

3 Data Exploration and Preprocessing

3.1 Initial Exploration

The first step was to load the data using the provided column file so that each raw value could be assigned to the correct variable name. I then reviewed the overall shape of the dataset, feature types, label distribution, and missing-value patterns.

This initial inspection highlighted two practical issues. First, the income classes were strongly imbalanced. Second, several migration-related and birthplace-related fields contained many missing values represented by the placeholder “?”.

3.2 Handling Missing Values

The placeholder “?” was converted to null values so the fields could be handled consistently during preprocessing. I chose imputation instead of dropping incomplete rows, because removing all observations with missing values would have unnecessarily reduced the available training data.

For numeric variables, I used median imputation. For categorical variables, I used the most frequent category. This is a simple and stable strategy for a mixed-type dataset of this size.

3.3 Target and Weight Treatment

For the classification task, the income label was converted into a binary target:

- 0 = income at or below $50,000
- 1 = income above $50,000

The survey weight was separated from the feature set and used as a sample weight during classifier training. This was done to better reflect the population representation described in the project brief.

For the segmentation task, I excluded both the income label and the survey weight from the clustering inputs. The label was excluded because segmentation should not be built directly from the answer variable. The weight was excluded because it reflects survey design rather than a direct demographic or behavioral characteristic of the individual.

3.4 Feature Preparation

The dataset contains both numeric and categorical features, so a single preprocessing approach would not have been appropriate. I used a pipeline with the following steps:

- numeric features: median imputation followed by standardization.
- categorical features: most-frequent imputation followed by one-hot encoding.

This ensured that all inputs were converted into a consistent machine-learning-ready format.

4 Classification Approach

4.1 Modeling Strategy

For the classification task, I compared two models:

1. **Logistic Regression** as a strong and interpretable baseline
2. **Random Forest** as a more flexible nonlinear alternative

This comparison was intentional. Logistic Regression is easier to explain and often performs well on structured tabular data after proper preprocessing. Random Forest can capture nonlinear relationships and interactions, which makes it a useful benchmark when evaluating whether added complexity improves results.

4.2 Training Procedure

I split the cleaned data into training and test sets using an 80/20 split with stratification on the income label. Stratification preserved the original class balance across both sets.

The preprocessing pipeline was applied within the modeling workflow so that missing-value handling, encoding, and scaling were performed consistently for both models. The survey weight was passed into model training as a sample weight.

4.3 Evaluation Procedure

Because of the class imbalance, I evaluated performance using several metrics rather than relying on accuracy alone:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC
- Confusion matrix

This gave a better view of how well the models identified the smaller higher-income group.

5 Classification Results

5.1 Test Set Performance

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.9522	0.7131	0.3845	0.4996	0.9449
Random Forest	0.9527	0.7273	0.3813	0.5003	0.9420

5.2 Interpretation

The two models performed very similarly. Random Forest was slightly stronger on precision and F1-score, while Logistic Regression was slightly stronger on ROC-AUC. In practical terms, both models were good at ranking individuals by likelihood of being in the higher-income group.

The main limitation was recall. At the default decision threshold, both models identified only a minority of the truly higher-income cases. This means the models are fairly conservative: when they predict that someone belongs to the higher-income class, they are often correct, but they still miss many such individuals.

From a business perspective, this makes the classifier more suitable as a prioritization tool than as a full capture tool. It can help identify a subset of likely higher-income prospects for premium

or selective campaigns, but it should not be interpreted as finding every potentially valuable customer.

5.3 Model Recommendation

I recommend **Logistic Regression** as the primary model for this project. The main reason is not that it clearly outperformed Random Forest, but that it delivered nearly identical results while remaining simpler to interpret and easier to communicate.

For a client-facing marketing application, that tradeoff is attractive. A more complex model is not automatically better if the performance gain is marginal.

One useful next step would be threshold tuning. Since both models achieved high ROC-AUC but modest recall, lowering the classification threshold below 0.5 would likely improve recall if the client wants broader higher-income outreach and is willing to accept lower precision.

6 Segmentation Approach

6.1 Modeling Strategy

For the segmentation task, I selected **K-Means** as the primary method. I chose it because it is scalable, easy to explain, and well suited to producing operational customer groups for marketing.

Because one-hot encoding creates a high-dimensional feature space, I applied **Principal Component Analysis (PCA)** before clustering. PCA reduced the feature set to 10 components, which helped make the clustering step more manageable and more stable.

6.2 Procedure

The segmentation workflow was:

1. remove “label” and “weight” from clustering inputs.
2. impute missing values.
3. scale numeric features and one-hot encode categorical features.
4. apply PCA
5. fit K-Means for multiple values of “k”
6. compare solutions using silhouette score.
7. fit the final clustering model and assign each record to a segment.
8. profile each segment using summary statistics and representative categories.

6.3 Cluster Selection

I tested values of “k” from 2 to 8. The resulting silhouette scores were:

Number of clusters (k)	Silhouette score
2	0.3472
3	0.3565
4	0.3559
5	0.3711
6	0.3752
7	0.3963
8	0.3935

The best score among the tested solutions was observed at **k = 7**, so I retained the 7-cluster solution for the main analysis.

7 Segmentation Results

The 7-cluster solution produced several segments that were clearly different from one another, along with a few that were relatively similar.

7.1 Cluster Profiles

Cluster 0: Children / Dependents / Non-Earners

- Average age: about 8 years
- Occupation: not in universe Greater-than-$50k
- rate: effectively 0%
- Size: 52,893

This segment appears to represent children or dependents who are not active earners. It is not directly useful for individual income-based targeting, although it may still matter indirectly in household-level strategy.

Cluster 4: Older Non-Working Adults

- Average age: about 56.5 years
- Wage per hour: 0
- Occupation: not in universe
- Greater-than-$50k
- rate: about 1.6%
- Size: 45,172

This segment appears to capture older adults with little workforce participation, such as retirees, homemakers, or otherwise non-working adults. From a marketing standpoint, this group may respond better to value-based, convenience-oriented, and household-essential offers.

Cluster 2: Affluent Investment-Heavy Segment

- Average age: about 46.8 years
- Extremely high capital gains
- Greater-than-$50k
- rate: about 88.3%
- Size: 392

This is the most economically distinctive segment in the dataset. Although small, it appears to represent affluent individuals with strong investment activity and a very high probability of belonging to the higher income group. This segment is a strong candidate for premium product promotion, loyalty campaigns, and convenience-focused messaging.

Cluster 5: Financially Active Asset-Holding Households

- Average age: about 44.0 years
- Very high capital losses
- Greater-than-$50k
- rate: about 30.5%
- Size: 3,794

This segment appears to represent financially active households, potentially including asset-holding or investment-engaged individuals. It is smaller than the mainstream population but potentially valuable for selective premium campaigns.

Clusters 1, 3, and 6: Broad Working-Age Mainstream Employed Adults

These three clusters shared several characteristics:

- average age roughly in the 30s to early 40s
- administrative or clerical occupation patterns
- low-to-moderate higher-income rates
- large population sizes

In business terms, these clusters likely represent mainstream employed adults. They appear well suited to broad retail campaigns, practical household offers, value-for-money promotions, and family-oriented messaging.

7.2 Practical Interpretation

One important finding is that the **best quantitative solution was not necessarily the simplest operational solution**. The 7-cluster model achieved the strongest silhouette score, but several of the resulting clusters were fairly similar in business terms.

That does not invalidate the model. It simply means that the analytical segmentation can be more detailed than the final marketing rollout needs to be. In practice, the client could retain the 7-cluster solution for analysis and insight generation while collapsing similar groups into a smaller number of campaign personas for day-to-day use.

This is the main business judgment behind my recommendation for Question 2.

8 Marketing Recommendations

The classification and segmentation outputs are most valuable when used together.

8.1 Use the classifier for prioritization

The classification model can be used to identify individuals who are more likely to belong to the higher income group. This is useful for premium targeting, higher margin offers, and campaigns where efficiency matters more than broad reach.

8.2 Use the segmentation model for message design

The segmentation model helps determine what type of message is likely to fit each customer group.

Examples:

- **Affluent investment-heavy segment**: premium products, loyalty benefits, convenience, exclusivity
- **Financially active asset-holding households**: selective premium bundles, value plus sophistication, financially oriented promotions
- **Mainstream working-age adults**: practical household offers, everyday value, family bundles, convenience pricing.
- **Older non-working adults**: reliability, affordability, convenience, essential categories
- **Children/dependents segment**: not a direct target, but useful for household composition insights

8.3 Combine both models

A practical workflow would be to first use the classifier to prioritize likely higher-income individuals and then use the segmentation model to tailor campaign creative, offer type, or channel strategy. This combined approach is stronger than using either model alone.

9 Limitations and Future Work

First, the dataset is based on historical census survey data from the mid-1990s, so the exact segment behavior may not translate directly to a modern retail environment.

Second, the income target is simplified into only two groups. That makes the classification problem manageable, but it also hides variation within each group.

Third, the clustering results depend on preprocessing choices, PCA, and the selected number of clusters. While the 7-cluster solution performed best by silhouette score, a smaller number of practical personas may be easier to operationalize.

Fourth, because the data includes demographic characteristics, any real-world use of these models should be reviewed carefully for fairness, bias, and compliance considerations.

Future improvements could include threshold tuning for classification, more detailed feature importance analysis, comparison with a mixed-type clustering method such as K-Prototypes, and segment stability testing.

10 Conclusion

This project produced two useful outputs from the census dataset.

The first is a classification model that helps identify individuals who are more likely to belong to the higher income group. Logistic Regression is recommended as the primary model because it combines strong predictive performance with simplicity and interpretability.

The second is a segmentation model that groups the population into meaningful profiles. The analysis identified several clear segments, including dependents, older non-working adults, affluent investment heavy individuals, financially active households, and large mainstream working-age groups.

Together, these tools give the client a practical framework for improving marketing efficiency, sharpening targeting, and designing more relevant customer communications.

References

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